

CH 5 Solution of Algebraic Equations

Throughout our discussion, we assume that

(i) the function $f(x)$ is continuous and continuously differentiable up to sufficient number of times.

(ii) $f(x) = 0$ has no multiple root, i.e., if α is a real root of $f(x) = 0$, in a sufficiently small interval (a, b) , then $f(\alpha) = 0$ and either $f'(x) < 0$ or $f'(x) > 0$ in (a, b) .

Algebraic Equations

An **algebraic equation** is an equation that consists of **only algebraic expressions**, meaning it involves **polynomials with rational coefficients** and uses operations like **addition, subtraction, multiplication, division, and exponentiation with integer exponents**.

✓ **General Form:**

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = 0$$

✓ **Examples:** Linear Equation, Quadratic Equation, Cubic Equation

Transcendental Equations

A **transcendental equation** is an equation that involves **transcendental functions**, meaning it includes functions that are **not algebraic**, such as **exponential, logarithmic, trigonometric, or other non-polynomial functions**.

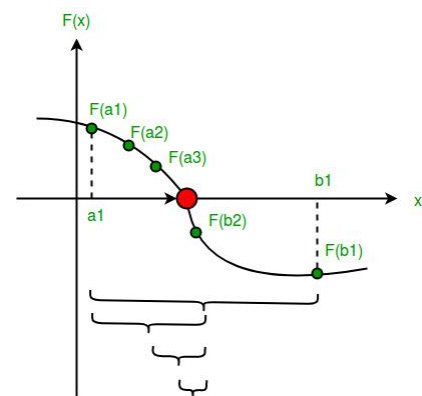
✓ **General Form:** $f(x)=g(x)$ (Either both or $f(x)$ or $g(x)$ are TE)

✓ **Examples:** Exponential Equation, Logarithmic Equation, Trigonometric Equation

Bisection Method

The **Bisection Method** is a root-finding algorithm that works by repeatedly dividing an interval $[a, b]$ in half and selecting the subinterval where the function changes sign. It guarantees convergence if the function is continuous.

- It is a **bracketing method**, meaning it requires an interval $[a, b]$ where $f(a)f(b)$ have opposite signs.
- The method repeatedly halves the interval and selects the subinterval containing the root.
- It converges **linearly** to the root.
- It always works if the function is continuous and the initial interval is correctly chosen.



✓ **Advantages:**

- ✓ **Guaranteed convergence** if the function is continuous and has a sign change.
- ✓ **Simple and easy to implement.**
- ✓ **Always finds a root within the interval.**

✗ **Disadvantages:**

- ✗ **Slow convergence** (linear rate).
- ✗ **Inefficient for highly accurate results** as it requires many iterations.
- ✗ **Cannot find multiple roots** in a given interval.

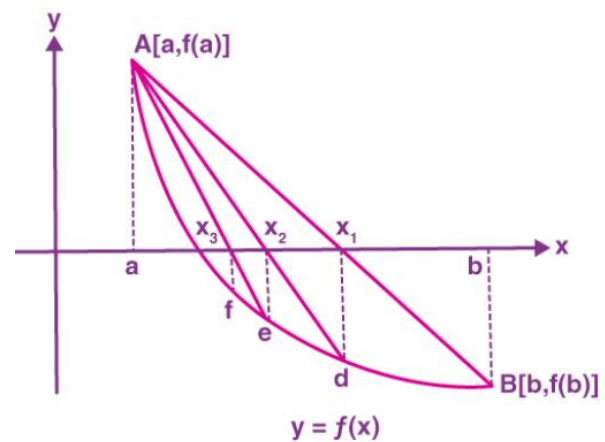
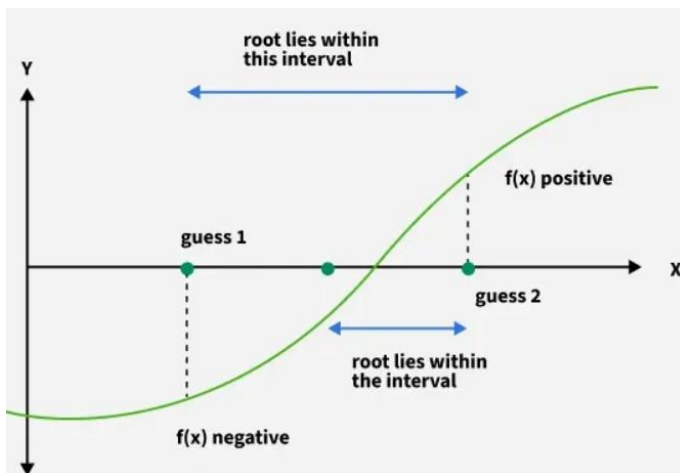
Regula Falsi Method (False Position Method)

The Regula Falsi Method is a root-finding algorithm that improves upon Bisection by using a secant line to approximate the root, instead of halving the interval. The new root estimate is found at:

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

where $f(a)$ and $f(b)$ have opposite signs.

- It is also a **bracketing method**, requiring an interval $[a, b]$ with opposite signs at endpoints.
- Instead of halving the interval, it takes a weighted average using a **secant line** to estimate the root.
- It converges **faster than the Bisection Method** but still slower than Newton-Raphson.
- The interval size does not always shrink symmetrically.



✓ Advantages:

- ✓ **Faster than the Bisection Method** (super-linear convergence).
- ✓ **Guaranteed to converge** if the function is continuous and the interval is chosen correctly.
- ✓ **More accurate root estimation** compared to the Bisection Method.

✗ Disadvantages:

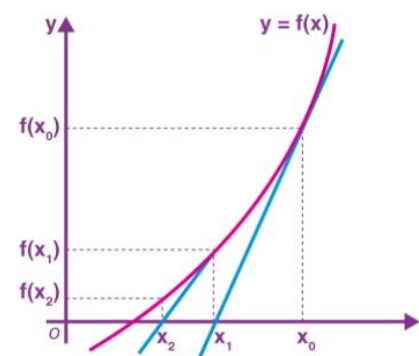
- ✗ **Convergence can be slow** in some cases when one end remains fixed for many iterations.
- ✗ **Not as fast as Newton-Raphson** for well-behaved functions.
- ✗ **Requires function evaluation at every step**, which increases computation time.

Newton-Raphson Method

The Newton-Raphson Method is an iterative root-finding technique that uses tangent lines to approximate the root. Given an initial guess x_0 , the next approximation is given by:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- It is an **open (non-bracketing) method**, meaning it starts with a single initial guess instead of an interval.
- It has **quadratic convergence**, making it much faster than both Bisection and Regula Falsi.
- Works well when the initial guess is close to the root.



✓ **Advantages:**

- ✓ **Very fast convergence** (quadratic rate).
- ✓ **Efficient for smooth, differentiable functions.**
- ✓ **Can handle multiple roots** if the function's behaviour is well understood.

✗ **Disadvantages:**

- ✗ **Requires computation of derivatives**, which may be difficult for some functions.
- ✗ **Fails if the derivative is zero or very small**, leading to division errors.
- ✗ **May diverge** if the initial guess is not chosen wisely.

Feature	Bisection Method	Regula Falsi Method	Newton-Raphson Method
Concept	Halves the interval at each step.	Uses a secant line to approximate the root.	Uses tangents to refine the root estimate.
Convergence Speed	Slow (linear convergence).	Faster than Bisection but may stagnate.	Very fast (quadratic convergence).
Formula Used	$c = \frac{a+b}{2}$	$c = \frac{af(b)-bf(a)}{f(b)-f(a)}$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
Initial Condition	Two values with opposite signs.	Two values with opposite signs.	One initial guess.
Derivative Needed?	✗ No	✗ No	✓ Yes
Guaranteed Convergence?	✓ Yes (if function is continuous).	✓ Yes (but slower in some cases).	✗ No (depends on good initial guess).
Efficiency	Least efficient.	Moderate efficiency.	Most efficient when applicable.
Best for?	Any continuous function with a sign change.	When secant approximation is effective.	When the function is smooth & derivative is available.